

VOLITION

Adaptive Biosignal Platform for Universal Assistive Control

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vivo Ignite 4th Edition (2026) · "Tech for Good"

Category: Community Development Innovations / HealthTech Innovations

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The unmet need

For people with severe motor impairment, the challenge is not only movement; it is reliable access to the world around them.

65 LAKH+
Estimate of Indians with motor disabilities (currently, based on Census 2011)

₹1–25+ LAKH
Usual commercial assistive-device cost range

ACCESS	High-cost assistive devices can be difficult to obtain.
ADAPTATION	Biosignals vary from person to person; fixed thresholds do not.
USABILITY	Fitting, recalibration and connectivity can add friction.

THE DESIGN QUESTION

Can the system learn the user, instead of forcing constant re-tuning?

This question defines the core design goal of Volition.

What did I build?

A working wearable + software system

NEXUS is the physical interface; Volition is the control platform around it.

CURRENT CAPABILITIES

01 EEG + EMG biosignal input

02 Adaptive intent detection

03 HID computer control

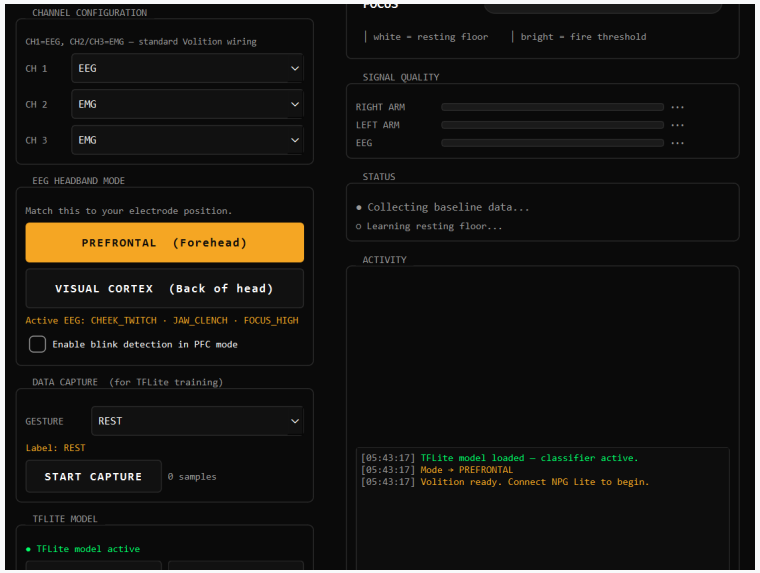
04 IMU-assisted spatial control

ONE PLATFORM · MANY CONTROLS



PHYSICAL PROTOTYPE

EEG + EMG · adjustable band · reusable contacts



VOLITION SOFTWARE

Live signal monitoring, intent decoding, HID configuration

Why this approach?

Existing approaches solve pieces; Volition is designed to combine them in one adaptable control platform.

	COMMERCIAL PROSTHETICS	CONSUMER EEG	DIY / OPEN BCI	VOLITION
AFFORDABILITY	High cost	Varies	Accessible	**Prototype-scale**
ADAPTATION	Clinical fitting	Limited	Often manual	**Adaptive baseline**
MULTI-MODAL	Device-specific	Varies	Flexible	**EEG + EMG**
CONTROL SCOPE	Device-specific	Limited	Flexible	**General-purpose**

MULTI-MODAL · ADAPTIVE · OFFLINE-FIRST · GENERAL-PURPOSE

Does it work?

The prototype was tested against four practical claims.

INTENT

Can deliberate signals trigger a defined action?

WHAT IT DEMONSTRATED

Demonstrated intent triggering

EEG / EMG → action

ADAPTATION

Can the system keep working as signal conditions change?

WHAT IT DEMONSTRATED

Adaptive baseline / reduced re-tuning

USER SIGNAL → ADAPTS

Direct Interfacing

Can intent control ordinary computer input?

WHAT IT DEMONSTRATED

Keyboard / HID control

INTENT → HID

Spatial Interfacing

Can intent extend to physical-device control?

WHAT IT DEMONSTRATED

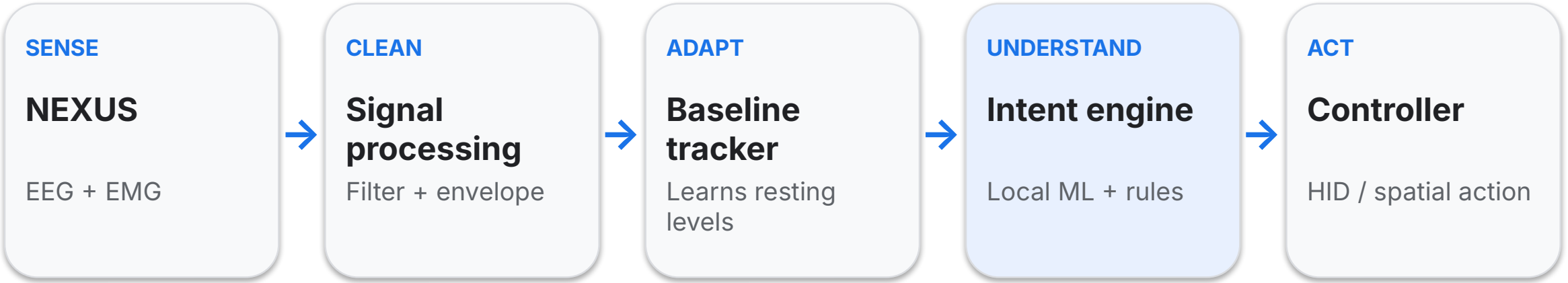
IMU-assisted spatial control

INTENT + DIRECTION → DEVICE

Prototype validation ≠ clinical validation. The next step is supervised human testing.

How Volition works

The implementation turns changing biosignals into a small, interpretable intent representation.



RAW SIGNAL → 9-VALUE VECTOR → LOCAL ML → INTENT → ACTION

Local inference · no recurring connection required

Core difference: the system adapts its interpretation to the user's signal.

Can it become practical?

The current prototype is low-cost to build; real-world use depends on validation, reliability and deployment.

AFFORDABILITY

~₹4,000–8,500

Current prototype build range

Open hardware · reusable contacts · offline-first

SCALABILITY

One control layer

NEXUS can connect to different outputs

Computer · physical devices · future phone app

REMAINING BARRIERS

Not yet deployment-ready

Signal quality · caregiver acceptance · edge cases

Clinical evidence is still required

NEXT STEP: supervised human validation before any clinical deployment claims.

From prototype to clinical validation

A clinical study / trial is a future validation step — not something completed by the current prototype.

01

ENGINEERING VALIDATION and IMPROVEMENT

Repeatability, false-trigger rate, signal robustness

02

SUPERVISED PILOT

Small participant group; usability + adaptation

03

CLINICAL STUDY / TRIAL

Safety, usability, performance across users

04

REAL-WORLD VALIDATION

Caregiver workflow + longer-term use

WHAT THE VALIDATION MUST ESTABLISH

- Safety
- Intent accuracy
- False-trigger rate
- Usability + adaptation

This pathway is the bridge from a working prototype to evidence suitable for future clinical adoption.

Let’s see it.

LIVE DEMO

Watch the complete loop:

01 BIOSIGNAL

NEXUS captures a deliberate signal

02 INTENT

Volition identifies the intended action

03 CONTROL

The computer / physical device responds

DEMO ORDER · INTENT → HID → SPATIAL CONTROL

The prototype is the evidence. The demo is where you, the jury, sees it.

Make technology adapt to the person - not the other way around.

ADAPTIVE

Learns changing user signals

MULTI-MODAL

Combines biosignals with orientation

ACCESSIBLE

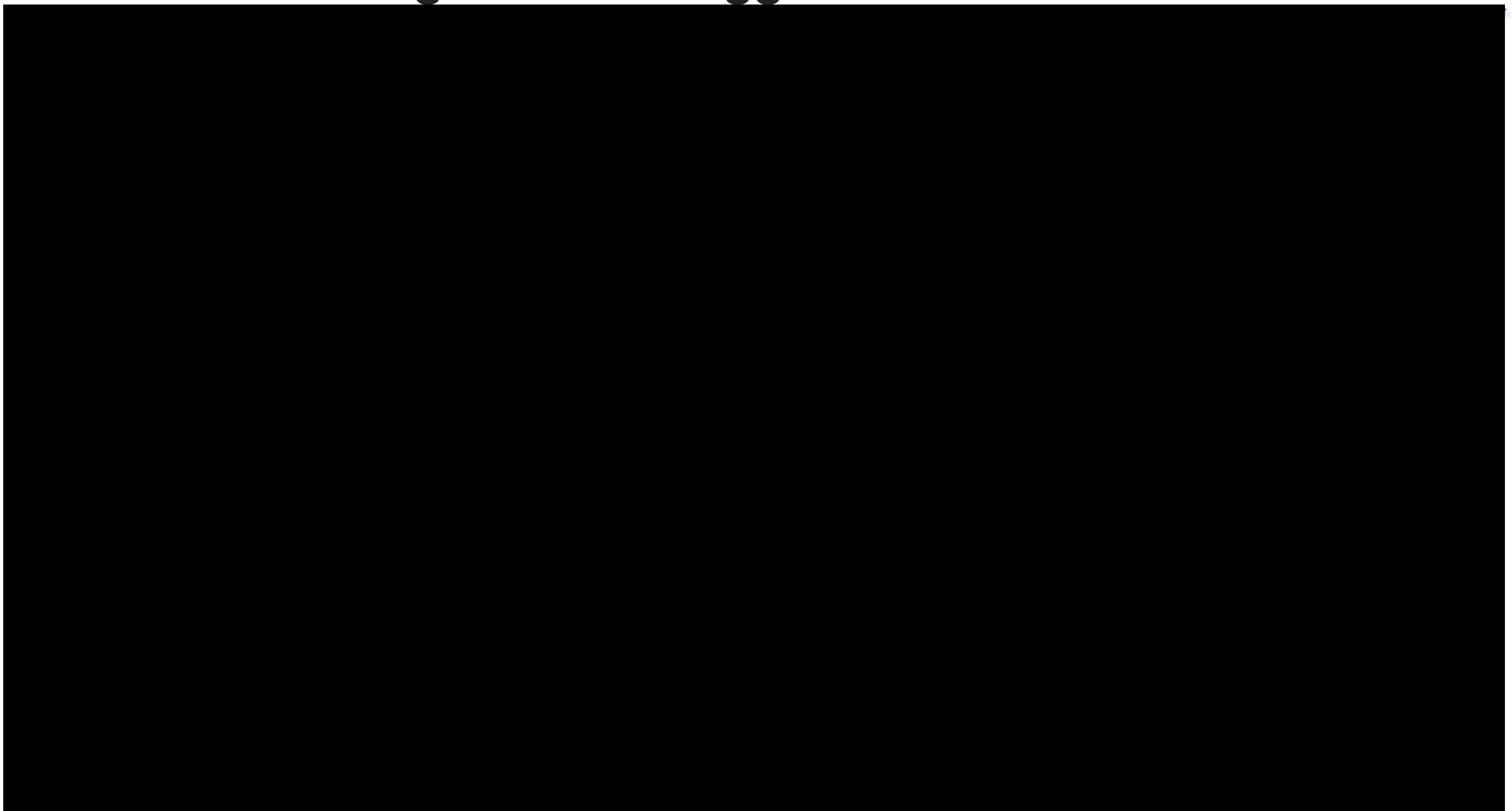
Prototype-scale cost; offline-first

Biosignal → intent → control

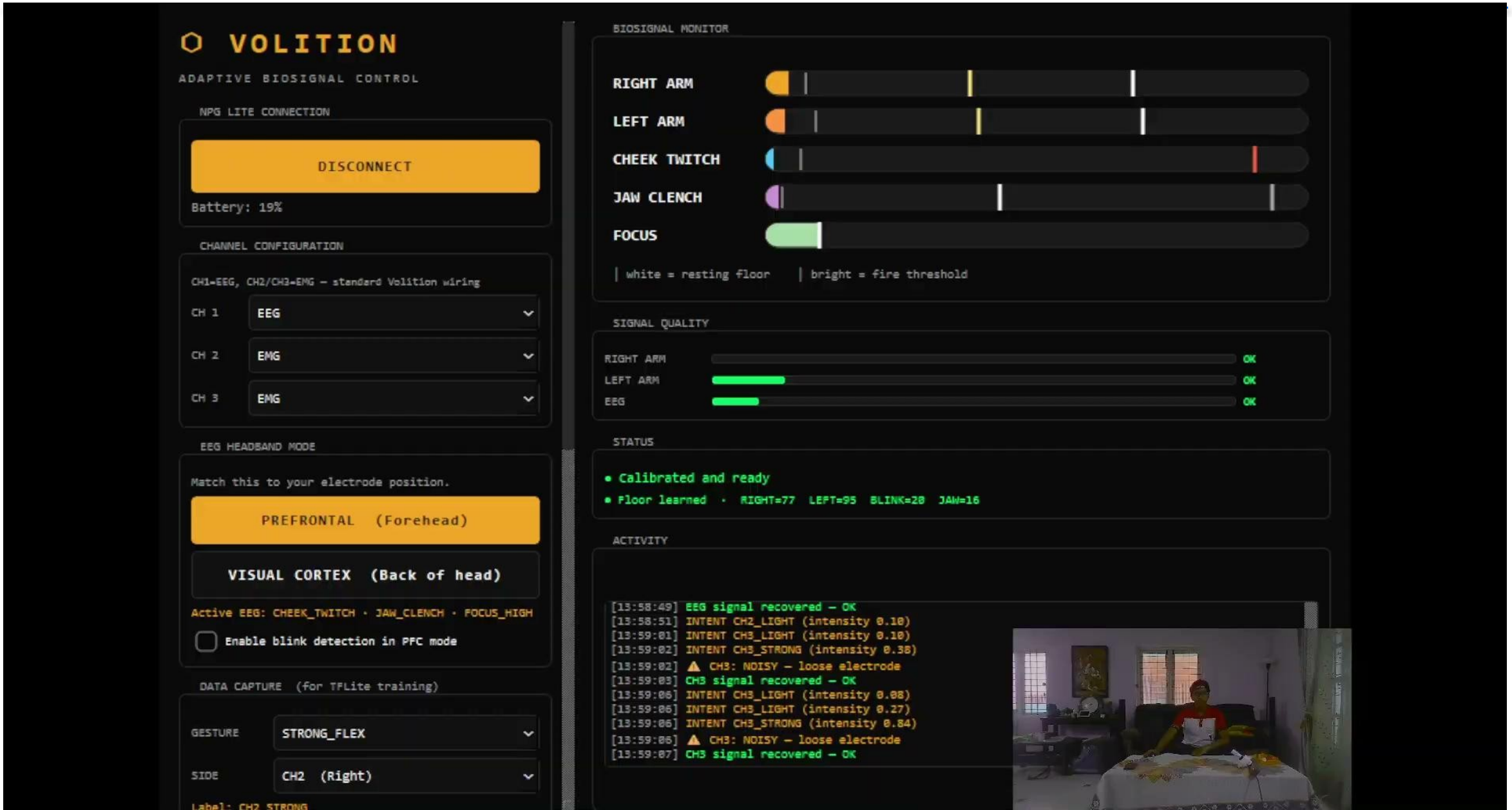
The current prototype proves the control concept; clinical validation is the next evidence layer.

Thank you.

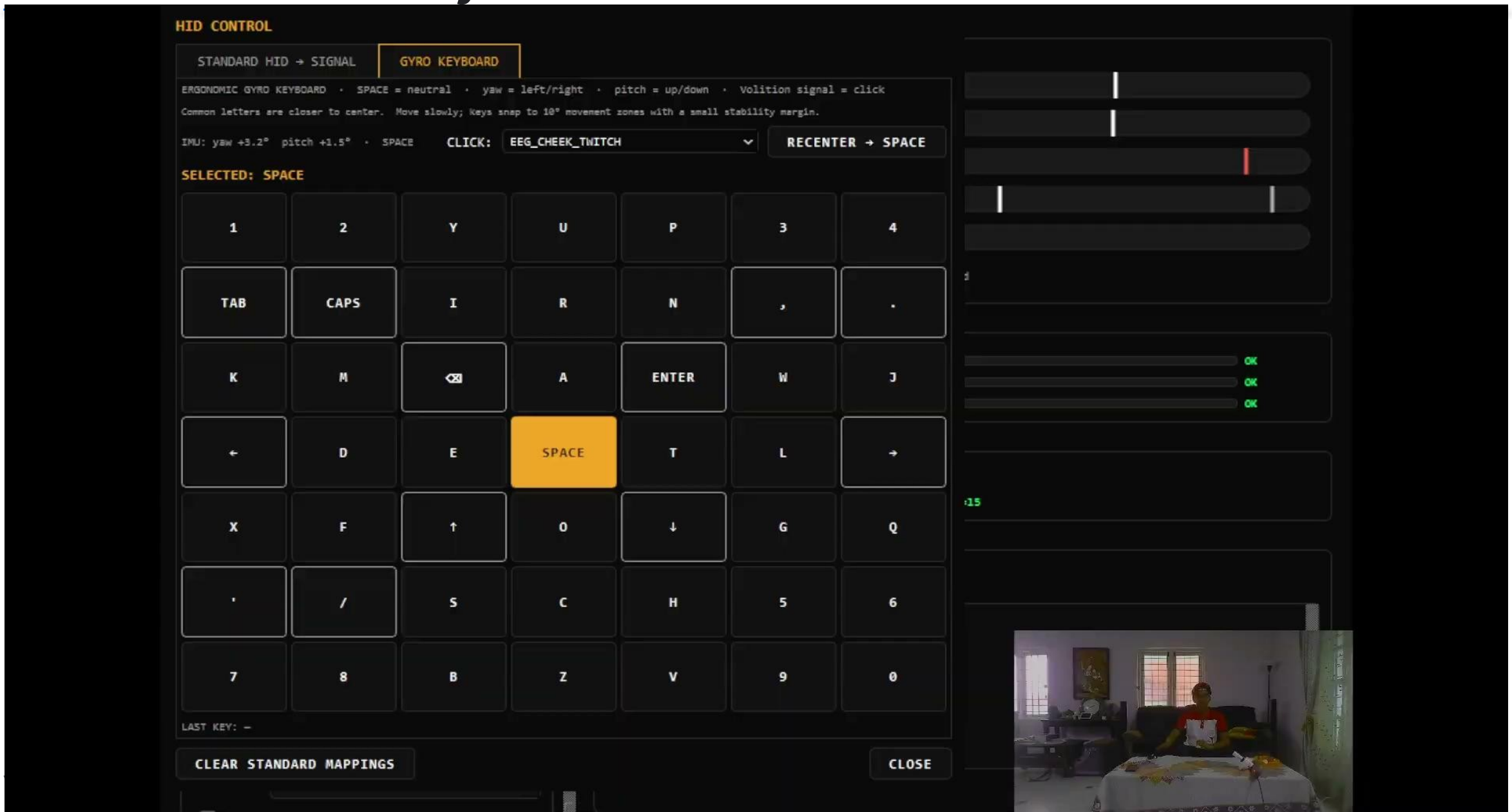
Let’s see it - Biosignal intent trigger



Let’s see it - Error detection



Let’s see it - HID Keyboard



Let’s see it - Spatial Triggering

